

Problem 11

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Problem 1 Verify that the given function satisfies the hypotheses of the Mean Value Theorem in the given interval. Then algebraically find all numbers c that satisfy the conclusion of the Mean Value Theorem.

$$f(x) = x + \sin\left(\frac{\pi}{4} \cdot x\right) \quad \text{on } [0, 2]$$

Explanation. The function f is continuous on $[0, 2]$ and differentiable on $(0, 2)$ since it is a sum of a linear polynomial and a composition of a sine function and another linear polynomial. Therefore, f satisfies the hypotheses of the Mean Value Theorem. We have that

$$f'(x) = 1 + \frac{\pi}{4} \cdot \cos\left(\frac{\pi}{4} \cdot x\right)$$

$$\frac{f(2) - f(0)}{2 - 0} = \frac{3}{2}$$

So we are looking to find all points $c \in (0, 2)$ which satisfy that $f'(c) = \frac{3}{2}$. So we solve:

$$1 + \frac{\pi}{4} \cdot \cos\left(\frac{\pi}{4} \cdot c\right) = \frac{3}{2}$$

Therefore,

$$\frac{\pi}{4} \cdot \cos\left(\frac{\pi}{4} \cdot c\right) = \frac{1}{2}$$

and

$$\cos\left(\frac{\pi}{4} \cdot c\right) = \frac{2}{\pi}$$

$$\frac{\pi}{4} \cdot c = \arccos\left(\frac{2}{\pi}\right)$$

$$c = \frac{4}{\pi} \cdot \arccos\left(\frac{2}{\pi}\right) \approx 1.12$$

Let's confirm this result with the graph.

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